STOICHEIA / EUCLID X.2–3

A measure in common.

Saved turn 8; 2 copies fitted. Original longer length: the square diagonal √2. The original square side is one length unit. Selected reading 1; independently selected printed source p.2.

Continue this experiment in the app · Save this exact record as CSV

The turn you saved

A · THE LONGER LENGTH



2 copies of B fitted

B · THE CURRENT MEASURE

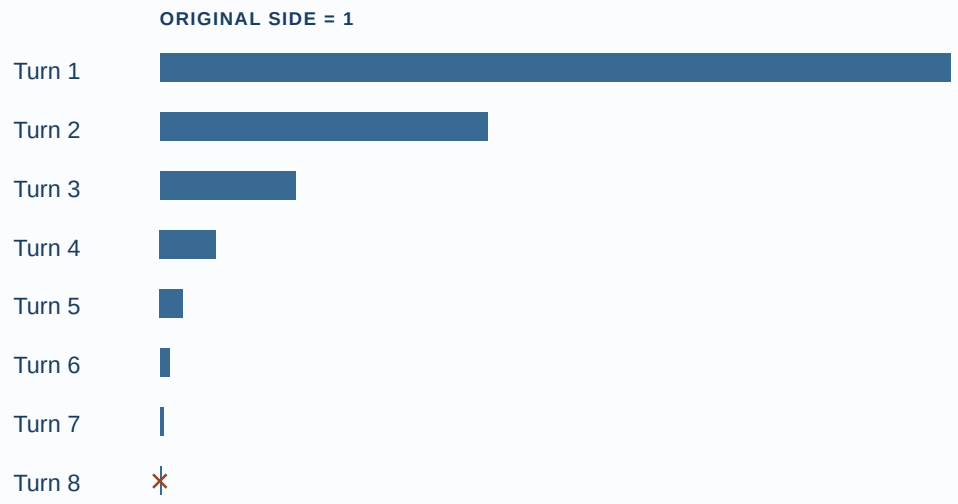


The positive remainder becomes the next measure ↓

The current uncut length is **577 − 408√2**. Local magnification makes the longer A fill the drawing at every turn. It does not keep the original ruler's scale.

This is an unfinished run. It has not established that the process never stops. The separate recurrence argument below establishes that condition for the diagonal.

All visited measures, on the original scale



A cross marks a positive length narrower than one SVG drawing unit. Its width is a locator, not the length. Exact expressions retain what a pixel cannot show.

Your subtraction record · exact lengths in original side units

Turn	Longer A	Measure B	Copies fitted	Left after those copies
1	√2	1	1	√2 − 1 (positive)
2	1	√2 − 1	2	3 − 2√2 (positive)
3	√2 − 1	3 − 2√2	2	5√2 − 7 (positive)
4	3 − 2√2	5√2 − 7	2	17 − 12√2 (positive)
5	5√2 − 7	17 − 12√2	2	29√2 − 41 (positive)
6	17 − 12√2	29√2 − 41	2	99 − 70√2 (positive)
7	29√2 − 41	99 − 70√2	2	169√2 − 239 (positive)
8	99 − 70√2	169√2 − 239	2	577 − 408√2 (positive)

Agreement through finitely many turns

At four decimals: diagonal 1.4142; fraction 1.4142. Their signed length difference (fraction − diagonal) is 2.123901e-6 original side units. These are distinct exact inputs, even if a ruler cannot resolve the difference.

The following separate comparison calculates both examples through your current turn, including its full division. It does not replace the partial copies in your saved record.

Two different exact lengths · computed comparison through turn 8

Turn	Square diagonal		Exact fraction 577/408	
	Copies	Remainder	Copies	Remainder
1	1	√2 − 1 > 0	1	169/408 > 0
2	2	3 − 2√2 > 0	2	35/204 > 0
3	2	5√2 − 7 > 0	2	29/408 > 0
4	2	17 − 12√2 > 0	2	1/34 > 0
5	2	29√2 − 41 > 0	2	5/408 > 0
6	2	99 − 70√2 > 0	2	1/204 > 0
7	2	169√2 − 239 > 0	2	1/408 > 0
8	2	577 − 408√2 > 0	2	0 = 0 · stops

A reason for every later diagonal turn

This is a modern specialization of the condition in X.2. Write k = √2 − 1, so 0 < k < 1. The first subtraction leaves k and produces the pair (1, k). Exactly two k fit into 1, because 1/k = 1 + √2 is strictly between 2 and 3. The remainder is 1 − 2k = 3 − 2√2 = k² > 0.

1



$k = \sqrt{2} - 1$



$k^2 = 3 - 2\sqrt{2}$

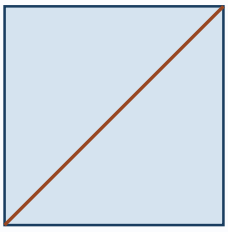


The new pair (k, k^2) equals $k \times (1, k)$. Multiplying both lengths by the same positive number preserves the copy count and gives a positive scaled remainder. The same argument applies again after any finite number of turns: zero is never reached. The finite experiment is limited to twenty-four turns; this identity supplies the infinite condition.

If a positive length E measured both originals, it would also measure every remainder, because a whole multiple can be subtracted without losing that property. Every nonzero grouped remainder is less than half the longer input: if $B \le A/2$, then $R < B \le A/2$; otherwise $q = 1$ and $R = A - B < A/2$. Repeated turns eventually leave a positive remainder smaller than E, which cannot contain even one E. X.2's contradiction therefore rules out the supposed common length measure.

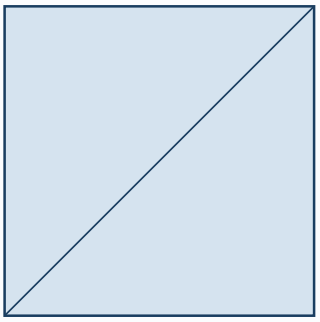
Yet the squares share an area measure

AREA = 1



side = 1

AREA = 2



side = √2

The squares on the side and diagonal have areas one and two in side-square units. They are commensurable in square, in the sense of Book X's second definition. This does not supply a common measure of the two lengths.

Read the selected passages

1. One measure for both

GREEK

Σύμμετρα μεγέθη λέγεται τὰ τῷ αὐτῷ μέτρῳ μετρούμενα, ἀσύμμετρα δέ, ὧν μηδὲν ἐνδέχεται κοινὸν μέτρον γενέσθαι.

HEIBERG'S LATIN · 1886

Magnitudines commensurabiles uocantur, quas eadem mensura metiri licet, incommensurabiles autem, quarum communis mensura inueniri nequit.

NEW ENGLISH READING

Magnitudes measured by the same measure are called commensurable; those for which no common measure can exist are called incommensurable.

σύμμετρα · commensurabiles · sharing a measure

τῷ αὐτῷ μέτρῳ is dative: 'by the same measure'. μετρούμενα is passive: 'being measured'. For these lengths, a measure fits a whole number of times, with no remainder. The starting unit need not itself be that measure.

Printed pp.2–3.

2. A different question about squares

GREEK

Εὐθεῖαι δυνάμει σύμμετροί εἰσιν, ὅταν τὰ ἀπ’ αὐτῶν τετράγωνα τῷ αὐτῷ χωρίῳ μετρῆται

HEIBERG’S LATIN · 1886

Rectae potentia commensurabiles sunt, ubi quadrata earum eadem mensura metiri licet

NEW ENGLISH READING

Straight lines are commensurable in square when the squares on them are measured by the same area.

δυνάμει · potentia · in square

In this definition, δύναμις concerns the square on a line. The diagonal and side of a square have no common length measure, yet their squares have areas 2 and 1 in side-square units. Do not carry the conclusion about lengths over to areas. A further word trap appears in definition 3 on the same pages: ῥητή / rationalis includes a line commensurable only in square with the chosen reference. Thus the diagonal can be ‘rational’ in that Book X classification while √2 is not a rational number in modern usage.

Printed pp.2–3.

X.2 · hypothesis, first clause

3. Keep subtracting in turn

GREEK

Ἐὰν δύο μεγεθῶν [ἐκκειμένων] ἀνίσων ἀνθυφαιρουμένου ἀεὶ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος

HEIBERG'S LATIN · 1886

Si ex duabus magnitudinibus inaequalibus minore semper uicissim a maiore subtracta

NEW ENGLISH READING

If, for two unequal magnitudes [set out], the smaller is always subtracted in turn from the greater…

ἀνθυφαιρουμένου · uicissim … subtracta · being subtracted in turn

The Greek uses a genitive absolute. ἐλάσσονος and μείζονος mean 'smaller' and 'greater'; their roles change after the turn. Latin uicissim means 'in turn'. The brackets around ἐκκειμένων belong to Heiberg's edition; they are not an extra condition added by this app.

Printed pp.6–7.

4. Never means never

GREEK

τὸ καταλειπόμενον μηδέποτε καταμετρῇ τὸ πρὸ ἑαυτοῦ, ἀσύμμετρα ἔσται τὰ μεγέθη.

HEIBERG'S LATIN · 1886

reliquum nunquam praecedentem magnitudinem metitur, magnitudines incommensurabiles erunt.

NEW ENGLISH READING

…what remains never measures the magnitude before it, then the magnitudes will be incommensurable.

μηδέποτε · nunquam · never

This is a condition on the entire process, not 'not yet after twenty-four stages'. καταμετρῇ means measures exactly. A finite run can find a final measure; an unfinished run alone cannot establish that there is none.

Printed pp.6–7.

X.2 · beginning of the contradiction

5. Suppose there were a measure

GREEK

Εἰ γάρ ἐστι σύμμετρα, μετρήσει τι αὐτὰ μέγεθος.

HEIBERG'S LATIN · 1886

Nam si commensurabiles sunt, magnitudo aliqua eas metietur.

NEW ENGLISH READING

For if they are commensurable, some magnitude will measure them.

μετρήσει · metietur · will measure

The proof temporarily assumes a common measure, named E in the next sentence. It does not assign E a numerical size. If E measures both lengths, it also measures any whole-number multiple and the remainder after subtracting one.

Printed pp.6–7.

6. Smaller than the supposed measure

GREEK

καὶ τοῦτο ἀεὶ γινέσθω, ἕως οὗ λειφθῇ τι μέγεθος, ὅ ἐστιν ἔλασσον τοῦ Ε.

HEIBERG'S LATIN · 1886

et hoc semper fiat, donec relinquatur magnitudo minor magnitudine E.

NEW ENGLISH READING

And let this keep happening until some magnitude is left that is smaller than E.

ἕως οὗ · donec · until

This supplies the size comparison needed by the contradiction. In our modern grouped divisions, every nonzero remainder is less than half the larger input: either the smaller input is at most half, or one subtraction already passes half. Repeated turns therefore fall below any supposed positive E. This explanation expands a step of the printed argument.

Printed pp.8–9.

7. The greater cannot measure the less

GREEK

καὶ λοιπὸν ἄρα τὸ ΑΗ μετρήσει, τὸ μεῖζον τὸ ἔλασσον· ὅπερ ἐστὶν ἀδύνατον.

HEIBERG'S LATIN · 1886

quare etiam reliquam AH metietur, maior minorem: quod fieri non potest.

NEW ENGLISH READING

So it will also measure the remainder AH—the greater measuring the less—which is impossible.

ἀδύνατον · fieri non potest · impossible

The subject is still the supposed measure E, carried through the preceding subtractions. A positive length smaller than E cannot contain even one whole E. The drawing's shrinking remainder illustrates the contradiction; its pixel size does not prove it.

Printed pp.8–9.

X.2 · conclusion

8. No common measure

GREEK

οὐκ ἄρα τὰ ΑΒ, ΓΔ μεγέθη μετρήσει τι μέγεθος· ἀσύμμετρα ἄρα ἐστὶ τὰ ΑΒ, ΓΔ μεγέθη.

HEIBERG'S LATIN · 1886

itaque magnitudines AB, ΓΔ nulla magnitudo metietur. ergo magnitudines AB, ΓΔ incommensurabiles erunt [def. 1].

NEW ENGLISH READING

Therefore no magnitude will measure the magnitudes AB and ΓΔ; therefore AB and ΓΔ are incommensurable.

ἄρα · ergo · therefore

The conclusion returns to the first definition. A display reading 0.000000 is not this conclusion: the remainder may be positive. The exact record keeps symbolic lengths separate from rounded decimal readings.

Printed pp.8–9.

X.3 · problem statement

9. When the process does stop

GREEK

Δύο μεγεθῶν συμμέτρων δοθέντων τὸ μέγιστον αὐτῶν κοινὸν μέτρον εὑρεῖν.

HEIBERG'S LATIN · 1886

Datis duabus magnitudinibus commensurabilibus maximam earum mensuram communem inuenire.

NEW ENGLISH READING

Given two commensurable magnitudes, to find their greatest common measure.

μέγιστον · maximam · greatest

δοθέντων says 'given'; εὑρεῖν is an infinitive stating the task. The last nonzero remainder measures every preceding length by working backward. Any other common measure must measure each remainder and hence the last one, so it cannot be greater. The complete printed proof continues through pp.10–13.

Printed pp.8–9.

X · first definitions · 1, final clause

10. Read the definition again, at the end

GREEK

ἀσύμμετρα δέ, ᾧν μηδὲν ἐνδέχεται κοινὸν μέτρον γενέσθαι.

HEIBERG'S LATIN · 1886

incommensurabiles autem, quarum communis mensura inueniri nequit.

NEW ENGLISH READING

And incommensurable are those for which no common measure can exist.

κοινόν · communis · common

'No common measure' does not mean 'no length', 'unmeasurable with a ruler', or 'not a real number'. In this workshop the diagonal has a definite length; what fails is a shared unit that fits both it and the side an exact whole number of times.

Printed pp.2–3.

Source and model boundaries

The printed X.2 concerns two magnitudes in general. The square diagonal, the rational examples, normalized magnifier, symbolic √2 notation and self-similarity specialization below are modern teaching choices. No discovery story or ancient use of these numerical examples is claimed. Heiberg's Latin is an 1886 translation, not an ancient witness. English and grammar notes are new reading aids.

J. L. Heiberg, *Euclidis opera omnia*, vol.III, Leipzig: B. G. Teubner, 1886, pp.2–13. Complete source PDF. Definition I.1 and the opening clause of I.2, selected clauses from X.2, and X.3's statement appear in the reading. Full scans include X.1 and the continuing proof of X.3. Handwriting in the scanned copy is not part of the typeset text.

Line wrapping and end-of-line hyphenation are removed; geometric labels use Greek Unicode. Heiberg's Latin u/v spelling and the Greek editorial brackets remain. Apparatus and inline reference markers omitted in selections remain in the whole-page images. English translations and grammar notes have not had independent philological peer review.

All subtraction counts, signs and zero decisions use exact BigInt arithmetic on (a + b√2)/d. Decimal values and SVG positions are rounded. The export records completed earlier turns and the actual partial current turn, without

inventing further performed operations. The comparison table and recurrence proof are separate calculations. No physical instrument tolerance or ancient measurement accuracy is modeled.

This HTML embeds its twelve pages and local font; it has no scripts or network subresources. External links need a connection. Print formatting and PDF inspection do not establish a physical print’s dimensions or accessibility tagging.

ι΄.

Ὅροι.

α΄. Σύμμετρα μεγέθη λέγεται τὰ τῷ αὐτῷ μέτρῳ μετρούμενα, ἀσύμμετρα δέ, ὧν μηδὲν ἐνδέχεται κοινὸν μέτρον γενέσθαι.

β΄. Εὐθεῖαι δυνάμει σύμμετροί εἰσιν, ὅταν τὰ ἀπ' αὐτῶν τετράγωνα τῷ αὐτῷ χωρίῳ μετρῆται, ἀσύμμετροι δέ, ὅταν τοῖς ἀπ' αὐτῶν τετραγώνοις μηδὲν ἐνδέχηται χωρίον κοινὸν μέτρον γενέσθαι.

γ΄. Τούτων ὑποκειμένων δείκνυται, ὅτι τῇ προτεθείσῃ εὐθείᾳ ὑπάρχουσιν εὐθεῖαι πλήθει ἄπειροι σύμμετροί τε καὶ ἀσύμμετροι αἱ μὲν μήκει μόνον, αἱ δὲ καὶ δυνάμει. καλείσθω οὖν ἡ μὲν προτεθεῖσα εὐθεῖα ῥητή, καὶ αἱ ταύτῃ σύμμετροι εἴτε μήκει καὶ δυνάμει εἴτε δυνάμει μόνον ῥηταί, αἱ δὲ ταύτῃ ἀσύμμετροι ἄλογοι καλείσθωσαν.

δ΄. Καὶ τὸ μὲν ἀπὸ τῆς προτεθείσης εὐθείας τετράγωνον ῥητόν, καὶ τὰ τούτῳ σύμμετρα ῥητά, τὰ δὲ τούτῳ ἀσύμμετρα ἄλογα καλείσθω, καὶ αἱ δυνάμεναι

Ad deff. cfr. Hero deff. 128—129, Anonymus Hultschii p. 256, Martianus Capella VI, 718.

Εὐκλείδου στοιχείων ῑ PV, *Εὐκλείδου στοιχείων τῆς θέωνος ἐκδόσεως ῑ* F, *Εὐκλείδου στοιχείων ῑ τῆς θέωνος ἐκδόσεως* b. 1. *ὅροι*] om. PFV, *ὅροι τοῦ ῑ* b, *ὅρος τοῦ ῑ* B. numeros om. codd. 5. Ante *σύμμετροι* ras. 1 litt P. 8. *ἐνδέχεται* bφ. 9. *προστεθείσῃ* b et e corr. F. 10. Post *εὐθεία* add. Theon: *τουτέστιν ἀφ' ἧς θέσει τὰ μέτρα τό τε πηχυαῖον καὶ τὸ παλαιστιαῖον καὶ τὸ δακτυλιαῖον ἢ τὸ ποδιαῖον λαμβάνεται* (BFVb).

Liber X.

Definitiones.

1. Magnitudines commensurabiles uocantur, quas eadem mensura metiri licet, incommensurabiles autem, quarum communis mensura inueniri nequit.

2. Rectae potentia commensurabiles sunt, ubi quadrata earum eadem mensura metiri licet, incommensurabiles autem, ubi nullum spatium communis quadratorum earum mensura inueniri potest.

3. His suppositis demonstratur, rectas numero infinitas esse datae rectae commensurabiles et incommensurabiles partim longitudine tantum, partim potentia quoque. iam data recta rationalis uocetur, et quae ei commensurabiles sunt siue longitudine potentiaque siue potentia tantum, rationales, quae autem ei incommensurabiles sunt, irrationales uocentur.

4. Et quadratum datae rectae rationale uocetur, et quae ei commensurabilia sunt, rationalia, quae autem ei incommensurabilia sunt, irrationalia, et rectae, quae

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πλήθει] om. F. *σύμμετροί τε καί*] supra scr. m. rec. P. 11. *μόνον, αἱ δὲ*] om. Theon (BFVb). 12. Post *δυνάμει* add. Theon: *αἱ δὲ δυνάμει μόνον* (BFVb). *προστεθεῖσα* b et e corr. F. 14. *σύμμετροι* b, corr. m. rec.; deinde add. Theon: *κατὰ τὸ συναμφότερον* (*συν-* om. b), *τουτέστιν* (*καὶ* del. F) *μήκει καὶ δυνάμει* (BFVb); idem P mg. m. 1 pro scholio. 16. *προστεθείσης* b et e corr. F. 17. *ῥητά*] om. F. 18. Ante *ἄλογα* add. *κατὰ τὸ συναμφότερον* F; idem P mg. m. 1 pro scholio. *καλείσθωσαν* Theon (BFVb).

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αὐτὰ ἄλογοι, εἰ μὲν τετράγωνα εἴη, αὐταὶ αἱ πλευραί, εἰ δὲ ἕτερά τινα εὐθύγραμμα, αἱ ἴσα αὐτοῖς τετράγωνα ἀναγράφουσαι.

α΄.

Δύο μεγεθῶν ἀνίσων ἐκκειμένων, ἐὰν ἀπὸ τοῦ μείζονος ἀφαιρεθῇ μεῖζον ἢ τὸ ἥμισυ καὶ τοῦ καταλειπομένου μεῖζον ἢ τὸ ἥμισυ, καὶ τοῦτο ἀεὶ γίγνηται, λειφθήσεταί τι μέγεθος, ὃ ἔσται ἔλασσον τοῦ ἐκκειμένου ἐλάσσονος μεγέθους.

Ἔστω δύο μεγέθη ἄνισα τὰ AB, Γ, ὧν μεῖζον τὸ AB· λέγω, ὅτι, ἐαν ἀπὸ τοῦ AB ἀφαιρεθῇ μεῖζον ἢ τὸ ἥμισυ καὶ τοῦ καταλειπομένου μεῖζον ἢ τὸ ἥμισυ, καὶ τοῦτο ἀεὶ γίγνηται, λειφθήσεταί τι μέγεθος, ὃ ἔσται ἔλασσον τοῦ Γ μεγέθους.

Τὸ Γ γὰρ πολλαπλασιαζόμενον ἔσται ποτὲ τοῦ AB μεῖζον. πεπολλαπαλσιάσθω, καὶ ἔστω τὸ ΔE τοῦ μὲν Γ πολλαπλάσιον, τοῦ δὲ AB μεῖζον, καὶ διῃρήσθω τὸ ΔE εἰς τὰ τῷ Γ ἴσα τὰ ΔZ, ZH, HE, καὶ ἀφῃρήσθω ἀπὸ μὲν τοῦ AB μεῖζον ἢ τὸ ἥμισυ τὸ $B\Theta$, ἀπὸ δὲ τοῦ $A\Theta$ μεῖζον ἢ τὸ ἥμισυ τὸ ΘK, καὶ τοῦτο ἀεὶ γιγνέσθω, ἕως ἂν αἱ ἐν τῷ AB διαιρέσεις ἰσοπληθεῖς γένωνται ταῖς ἐν τῷ ΔE διαιρέσεσιν.

Ἔστωσαν οὖν αἱ AK, $K\Theta$, ΘB διαιρέσεις ἰσοπληθεῖς οὖσαι ταῖς ΔZ, ZH, HE· καὶ ἐπεὶ μεῖζόν ἐστι τὸ ΔE τοῦ AB, καὶ ἀφῄρηται ἀπὸ μὲν τοῦ ΔE ἔλασσον τοῦ ἡμίσεως τὸ EH, ἀπὸ δὲ τοῦ AB μεῖζον ἢ τὸ ἥμισυ

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1. ἄλογα V, corr. m. 2. Deinde add. καλείσθωσαν Theon (BFVb). 2. ἴσαι φ. 5. ἐκκειμένων] ante ἀνίσων add. B mg. m. 1. 8. ἀεί] αἰεί F, ἀεὶ ἄν V? γίνηται V (η e corr.), γίγνεται b. ληφθήσεται Vb. 9. ἐστιν Theon (BFVb).

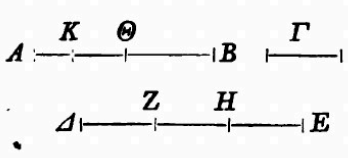
quadratae iis aequales sunt, irrationales uocentur, in quadratis ipsa latera, in ceteris figuris rectilineis eae, ex quibus quadrata illis aequalia construi possunt.

I.

Propositis duabus magnitudinibus inaequalibus si a maiore plus quam dimidium subtrahitur et a reliqua plus quam dimidium, et hoc semper fit, magnitudo relinquetur, quae minor erit proposita magnitudine minore.

Sint duae magnitudines inaequales AB, Γ, quarum maior sit AB. dico, si ab AB plus quam dimidium subtrahatur et ab reliqua plus quam dimidium, et hoc semper fiat, magnitudinem relictum iri, quae minor sit magnitudine Γ.

Nam Γ multiplicata aliquando magnitudine AB maior erit [cfr. V def. 4]. multiplicetur et ΔE magnitudinis Γ multiplex sit, eadem autem $> AB$, et ΔE in partes magnitudini Γ aequales ΔZ, ZH, HE diuidatur, et ab AB plus quam dimidium subtrahatur $B\Theta$, ab $A\Theta$ autem plus quam dimidium ΘK, et hoc semper fiat, donec in AB totidem diuisiones fiant, quot in ΔE.



ἔλαττον F. τοῦ] om. V? ἐγκειμένου b. ἐλάττονος F. 12. δὴ ὅτι b. 13. καὶ — ἥμισυ] om. P. καί] (prius) καὶ ἀπὸ V. 14. αἰεί F. γίγνεται V, γίνηται b. ληφθήσεται V. ἐστιν V. ἔλαττον F. 16. γάρ] ἄρα F. AB μεγέθους Theon (BFVb). 19. εἰς] m. rec. B. ἀπό] om. V. 21. γινέσθω P. 23. ταῖς] corr. ex ται m. rec. b. 24. οὖν] om. b. διαιρέσις P, sed corr. 25. HZ F. ἐστιν F. 26. τοῦ] (alt.) post ins. m. 1 F. 27. ἡμίσεος b, ἡμίσους V. τό] corr. ex τοῦ F. ἢ τὸ ἥμισυ] τοῦ ἡμίσεως F, τοῦ ἡμίσεος BVb.

τὸ $B\Theta$, λοιπὸν ἄρα τὸ $H\Delta$ λοιποῦ τοῦ ΘA μεῖζόν ἐστιν. καὶ ἐπεὶ μεῖζόν ἐστι τὸ $H\Delta$ τοῦ ΘA, καὶ ἀφῄρηται τοῦ μὲν $H\Delta$ ἥμισυ τὸ HZ, τοῦ δὲ ΘA μεῖζον ἢ τὸ ἥμισυ τὸ ΘK, λοιπὸν ἄρα τὸ ΔZ λοιποῦ τοῦ AK μεῖζόν ἐστιν. ἴσον δὲ τὸ ΔZ τῷ Γ· καὶ τὸ Γ ἄρα τοῦ AK μεῖζόν ἐστιν. ἔλασσον ἄρα τὸ AK τοῦ Γ.

Καταλείπεται ἄρα ἀπὸ τοῦ AB μεγέθους τὸ AK μέγεθος ἔλασσον ὂν τοῦ ἐκκειμένου ἐλάσσονος μεγέθους τοῦ Γ· ὅπερ ἔδει δεῖξαι. — ὁμοίως δὲ δειχθήσεται, κἂν ἡμίση ᾖ τὰ ἀφαιρούμενα.

β΄.

Ἐὰν δύο μεγεθῶν [ἐκκειμένων] ἀνίσων ἀνθυφαιρουμένου ἀεὶ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος τὸ καταλειπόμενον μηδέποτε καταμετρῇ τὸ πρὸ ἑαυτοῦ, ἀσύμμετρα ἔσται τὰ μεγέθη.

Δύο γὰρ μεγεθῶν ὄντων ἀνίσων τῶν AB, $\Gamma\Delta$ καὶ ἐλάσσονος τοῦ AB ἀνθυφαιρουμένου ἀεὶ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος τὸ περιλειπόμενον μηδέποτε καταμετρείτω τὸ πρὸ ἑαυτοῦ· λέγω, ὅτι ἀσύμμετρά ἐστι τὰ AB, $\Gamma\Delta$ μεγέθη.

Εἰ γάρ ἐστι σύμμετρα, μετρήσει τι αὐτὰ μέγεθος. μετρείτω, εἰ δυνατόν, καὶ ἔστω τὸ E· καὶ τὸ μὲν AB τὸ $Z\Delta$ καταμετροῦν λειπέτω ἑαυτοῦ ἔλασσον τὸ ΓZ,

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2. ἐστιν] comp. Fb, ἐστι BV. ἐστι] om. V. 4. η τὸ ἥμισυ] τοῦ ἡμίσεος BVb, τοῦ ἡμίσεως F. 7. καταλέλειπται Bb. 8. ἐγκειμένου b. ἐλάττονος F. 10. ἡμίσυη P, ἡμίσεα V. Seq. demonstr. altera, u. app. 12. ἐκκειμένων] mg. m. 1 P. ἀνθυφαιρομένου V, corr. m. 2. 13. αἰεί F. ἐλάττονος F. 15. τά] τό F, corr. m. 2. 16. καὶ ὄντος Theon (BFVb). 17. ἐλάττονος F. ἀνθυφαιρομένου V, corr. m. 2. αἰεί F. 19. ἐστιν P. 21. ἐστι] supra scr. -αι V. τι] om. F. 23.

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diuisiones igitur AK, $K\Theta$, ΘB numero aequales sint diuisionibus ΔZ, ZH, HE. et quoniam $\Delta E > AB$, et a ΔE minus quam dimidium subtractum est EH, ab AB autem plus quam dimidium $B\Theta$, erit $H\Delta > \Theta A$. et quoniam $H\Delta > \Theta A$, et ab $H\Delta$ dimidium subtractum est HZ, a ΘA autem plus quam dimidium ΘK, erit $\Delta Z > AK$. uerum $\Delta Z = \Gamma$. quare etiam $\Gamma > AK$. ergo $AK < \Gamma$.

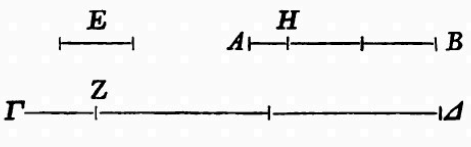
Ergo ex magnitudine AB relinquitur magnitudo AK minor proposita magnitudine minore Γ; quod erat demonstrandum.

Similiter autem demonstrabitur, etiam si, quae subtrahuntur, dimidia sunt.

II.

Si ex duabus magnitudinibus inaequalibus minore semper uicissim a maiore subtracta reliquum nunquam praecedentem magnitudinem metitur, magnitudines incommensurabiles erunt.

Datis enim duabus magnitudinibus inaequalibus AB, $\Gamma\Delta$ minor sit AB, et minore semper uicissim a maiore subtracta reliquum ne unquam praecedentem magnitudinem metiatur. dico, magnitudines AB, $\Gamma\Delta$ incommensurabiles esse.



Nam si commensurabiles sunt, magnitudo aliqua eas metietur. metiatur, si fieri potest, et sit E. et AB magnitudinem $Z\Delta$ metiens se ipsa minorem relinquat

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$Z\Delta$] mut. in $\Gamma\Delta$ m. 2 B, m. rec. b; ΔZ e corr. PV. ἐλάσσονα P, sed α del.

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τὸ δὲ ΓΖ τὸ ΒΗ καταμετροῦν λειπέτω ἑαυτοῦ ἔλασσον τὸ ΑΗ, καὶ τοῦτο ἀεὶ γινέσθω, ἕως οὗ λειφθῇ τι μέγεθος, ὅ ἐστιν ἔλασσον τοῦ Ε. γεγονέτω, καὶ λελείφθω τὸ ΑΗ ἔλασσον τοῦ Ε. ἐπεὶ οὖν τὸ Ε τὸ ΑΒ μετρεῖ, ἀλλὰ τὸ ΑΒ το ΔΖ μετρεῖ, καὶ τὸ Ε ἄρα τὸ ΖΔ μετρήσει. μετρεῖ δὲ καὶ ὅλον τὸ ΓΔ· καὶ λοιπὸν ἄρα τὸ ΓΖ μετρήσει. ἀλλὰ τὸ ΓΖ τὸ ΒΗ μετρεῖ· καὶ τὸ Ε ἄρα τὸ ΒΗ μετρεῖ. μετρεῖ δὲ καὶ ὅλον τὸ ΑΒ· καὶ λοιπὸν ἄρα τὸ ΑΗ μετρήσει, τὸ μεῖζον τὸ ἔλασσον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὰ ΑΒ, ΓΔ μεγέθη μετρήσει τι μέγεθος· ἀσύμμετρα ἄρα ἐστὶ τὰ ΑΒ, ΓΔ μεγέθη.

Ἐὰν ἄρα δύο μεγεθῶν ἀνίσων, καὶ τὰ ἑξῆς.

γ΄.

Δύο μεγεθῶν συμμέτρων δοθέντων τὸ μέγιστον αὐτῶν κοινὸν μέτρον εὑρεῖν.

Ἔστω τὰ δοθέντα δύο μεγέθη σύμμετρα τὰ ΑΒ, ΓΔ, ὧν ἔλασσον τὸ ΑΒ· δεῖ δὴ τῶν ΑΒ, ΓΔ τὸ μέγιστον κοινὸν μέτρον εὑρεῖν.

Τὸ ΑΒ γὰρ μέγεθος ἤτοι μετρεῖ τὸ ΓΔ ἢ οὔ. εἰ μὲν οὖν μετρεῖ, μετρεῖ δὲ καὶ ἑαυτό, τὸ ΑΒ ἄρα τῶν

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1. ΒΗ] in ras. P, mut. in ΒΑ B m. 2, in ΑΒ m. rec.; Η e corr V. 2. γιγνέσθω F. ληφθῇ BVb. 3. ἔσται P. ἔλαττον F. εἰλήφθω V. 4. τό] (pr.) τοῦ F. 5. ΖΔ P. ΖΔ] mut. in ΔΖ V, ΔΖ BFb. 8. ΒΗ] ΗΒ P. μετρεῖ] (prius) supra m. 2 F. 10. ἐστίν] om. V. 11. Post τι ras. 1 litt. V. ἐστίν P. 13. μεγεθῶν ἐκκειμένων F. καὶ τὰ ἑξῆς] ὅπερ ἔδει δεῖξαι V (post ἑξῆς add. Δ (π) b); ἐκκειμένων ἀνίσων· ἀνθυφαιρουμένου ἀεὶ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος τὸ καταλειπόμενον μηδέποτε καταμετρῇ τὸ πρὸ ἑαυτοῦ, ἀσύμμετρα ἔσται τὰ μεγέθη m. 2 V, del. ἀνίσων lin. 13. 17. ἔστωσαν F. σύμ-

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ΓZ, ΓZ autem BH metiens se ipsa minorem relinquat AH, et hoc semper fiat, donec relinquatur magnitudo minor magnitudine E. fiat et relinquatur $AH < E$. iam quoniam E magnitudinem AB metitur et AB magnitudinem ΔZ, etiam E magnitudinem $Z\Delta$ metitur. uerum etiam totam $\Gamma\Delta$ metitur. itaque etiam reliquam magnitudinem ΓZ metietur. sed ΓZ magnitudinem BH metitur. quare etiam E magnitudinem BH metitur. uerum etiam totam AB metitur. quare etiam reliquam AH metietur, maior minorem; quod fieri non potest. itaque magnitudines AB, $\Gamma\Delta$ nulla magnitudo metietur. ergo magnitudines AB, $\Gamma\Delta$ incommensurabiles erunt [def. 1].

Ergo si ex duabus magnitudinibus inaequalibus, et quae sequuntur.

III.

Datis duabus magnitudinibus commensurabilibus maximam earum mensuram communem inuenire.

Sint duae magnitudines datae commensurabiles AB, $\Gamma\Delta$, quarum minor sit AB. oportet igitur magnitudinum AB, $\Gamma\Delta$ maximam mensuram communem inuenire.

Nam magnitudo AB magnitudinem $\Gamma\Delta$ aut metitur aut non metitur. iam si metitur, et se ipsam quoque

---

μετρα μεγέθη V. 18. ἔλαττον F. 20. μέγεθος] om. Theon (BFVb). ἤτοι] m. rec. P. 21. Post οὖν add. τὸ AB τὸ $\Gamma\Delta$ V. μετρεῖ] (prius) supra m. 1 B. αὐτό B, corr. m. 2. τῶν AB, $\Gamma\Delta$] om. V.

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AB, *ΓΔ* κοινὸν μέτρον ἐστίν· καὶ φανερόν, ὅτι καὶ μέγιστον. μεῖζον γὰρ τοῦ *AB* μεγέθους τὸ *AB* οὐ μετρήσει.

Μὴ μετρείτω δὴ τὸ *AB* τὸ *ΓΔ*. καὶ ἀνθυφαιρουμένου ἀεὶ τοῦ ἐλάσσονος ἀπὸ τοῦ μείζονος, τὸ περιλειπόμενον μετρήσει ποτὲ τὸ πρὸ ἑαυτοῦ διὰ τὸ μὴ εἶναι ἀσύμμετρα τὰ *AB*, *ΓΔ*· καὶ τὸ μὲν *AB* τὸ *EΔ* καταμετροῦν λειπέτω ἑαυτοῦ ἔλασσον τὸ *EΓ*, τὸ δὲ *EΓ* τὸ *ZB* καταμετροῦν λειπέτω ἑαυτοῦ ἔλασσον τὸ *AZ*, τὸ δὲ *AZ* τὸ *ΓE* μετρείτω.

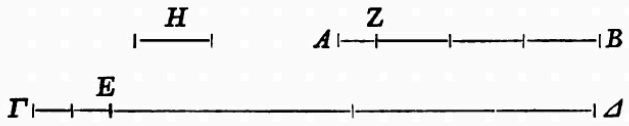
Ἐπεὶ οὖν τὸ *AZ* τὸ *ΓE* μετρεῖ, ἀλλὰ τὸ *ΓE* τὸ *ZB* μετρεῖ, καὶ τὸ *AZ* ἄρα τὸ *ZB* μετρήσει. μετρεῖ δὲ καὶ ἑαυτό· καὶ ὅλον ἄρα τὸ *AB* μετρήσει τὸ *AZ*. ἀλλὰ τὸ *AB* τὸ *ΔE* μετρεῖ· καὶ τὸ *AZ* ἄρα τὸ *EΔ* μετρήσει. μετρεῖ δὲ καὶ τὸ *ΓE*· καὶ ὅλον ἄρα τὸ *ΓΔ* μετρεῖ· τὸ *AZ* ἄρα τῶν *AB*, *ΓΔ* κοινὸν μέτρον ἐστίν. λέγω δή, ὅτι καὶ μέγιστον. εἰ γὰρ μή, ἔσται τι μέγεθος μεῖζον τοῦ *AZ*, ὃ μετρήσει τὰ *AB*, *ΓΔ*. ἔστω τὸ *H*. ἐπεὶ οὖν τὸ *H* τὸ *AB* μετρεῖ, ἀλλὰ τὸ *AB* τὸ *EΔ* μετρεῖ, καὶ τὸ *H* ἄρα τὸ *EΔ* μετρήσει. μετρεῖ δὲ καὶ ὅλον τὸ *ΓΔ*· καὶ λοιπὸν ἄρα τὸ *ΓE* μετρήσει τὸ *H*. ἀλλὰ τὸ *ΓE* τὸ *ZB* μετρεῖ· καὶ τὸ *H* ἄρα τὸ *ZB* μετρήσει. μετρεῖ δὲ καὶ ὅλον τὸ *AB*, καὶ λοιπὸν τὸ *AZ* μετρήσει, τὸ μεῖζον τὸ ἔλασσον· ὅπερ

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1. *ἐστίν*] comp. F, *ἐστί* Bb, *ἐστὶ τῶν AB, ΓΔ* V. *καί*] (alt.) *μέτρον ἐστί* V. 4. *καί*] om. BFVb. *ἀνθυφαιρομένου* V, sed corr. m. 2; *ἀνθυφαιρόμενον* F. 5. *ἀεί*] *ἄρα ἀεί* Vb, *ἄρα* F, om. B (*ἄρα ἀεί* m. 2). 8. *τὸ EΓ* — 9. *ἔλασσον*] m. 2 B. 10. *δὲ AZ*] *AZ δέ* P. 13. *μετρήσει* — 14. *AB*] mg. m. 1 P. 14. Post *AZ* ras. 1 litt. V. 16. *μετρεῖ*] *μετρησει* F. Deinde add. Theon: *τὸ AZ ἄρα τὰ AB, ΓΔ μετρεῖ* (BFVb); idem m. rec. P. *ἄρα*] om. *φ*. *ἐστί* BV, comp. Fb. 18. *τά*] *τό* B, corr. m. 2. Post *ΓΔ* add. *μετρείτω καί* V, sed punctis del. 20.

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metitur, AB magnitudinum AB, $\Gamma\Delta$ communis est mensura. et adparet, eandem maximam esse; nam magnitudo magnitudine AB maior AB non metietur.

itaque ne metiatur AB magnitudinem $\Gamma\Delta$. et minore semper uicissim a maiore subtracta reliquum aliquando magnitudinem praecedentem metietur, quia AB, $\Gamma\Delta$ incommensurabiles non sunt [cfr. prop. II]. et AB magnitudinem $E\Delta$ metiens se ipsa minorem relinquat $E\Gamma$, $E\Gamma$ autem ZB metiens se ipsa minorem relinquat AZ, et AZ magnitudinem ΓE metiatur. iam quoniam AZ magnitudinem ΓE metitur, ΓE autem ZB, etiam AZ magnitudinem ZB metietur. uerum etiam se ipsam metitur. quare etiam totam AB metietur AZ. sed AB magnitudinem ΔE metitur. itaque etiam AZ magnitudinem $E\Delta$ metietur. uerum etiam ΓE metitur. quare etiam totam $\Gamma\Delta$ metitur. itaque AZ magnitudinem AB, $\Gamma\Delta$ communis est mensura. iam dico, eandem maximam esse. nam si minus, magnitudo erit maior magnitudine AZ, quae AB, $\Gamma\Delta$ metiatur. sit H. iam quoniam H magnitudinem AB metitur, et AB magnitudinem $E\Delta$ metitur, etiam H magnitudinem $E\Delta$ metietur. uerum etiam totam $\Gamma\Delta$ metitur. quare etiam reliquam ΓE metitur H. sed ΓE magnitudinem ZB metitur. itaque etiam H magnitudinem ZB metitur. uerum etiam totam AB metitur et reliquam AZ me-

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$E\Delta$] (prius) ΔE P. 21. καί] (alt.) om. V. 23. τό] (alt.) τόν P. 24. λοιπὸν ἄρα F.

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ἐστὶν ἀδύνατον. οὐκ ἄρα μεῖζόν τι μέγεθος τοῦ AZ τὰ AB, $\Gamma\Delta$ μετρήσει· τὸ AZ ἄρα τῶν AB, $\Gamma\Delta$ τὸ μέγιστον κοινὸν μέτρον ἐστίν.

Δύο ἄρα μεγεθῶν συμμέτρων δοθέντων τῶν AB, $\Gamma\Delta$ τὸ μέγιστον κοινὸν μέτρον ηὕρηται· ὅπερ ἔδει δεῖξαι.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι, ἐὰν μέγεθος δύο μεγέθη μετρῇ, καὶ τὸ μέγιστον αὐτῶν κοινὸν μέτρον μετρήσει.

δ'.

Τριῶν μεγεθῶν συμμέτρων δοθέντων τὸ μέγιστον αὐτῶν κοινὸν μέτρον εὑρεῖν.

Ἔστω τὰ δοθέντα τρία μεγέθη σύμμετρα τὰ A, B, Γ· δεῖ δὴ τῶν A, B, Γ τὸ μέγιστον κοινὸν μέτρον εὑρεῖν.

Εἰλήφθω γὰρ δύο τῶν A, B τὸ μέγιστον κοινὸν μέτρον, καὶ ἔστω τὸ Δ· τὸ δὴ Δ τὸ Γ ἤτοι μετρεῖ ἢ οὔ [μετρεῖ]. μετρείτω πρότερον. ἐπεὶ οὖν τὸ Δ τὸ Γ μετρεῖ, μετρεῖ δὲ καὶ τὰ A, B, τὸ Δ ἄρα τὰ A, B, Γ μετρεῖ· τὸ Δ ἄρα τῶν A, B, Γ κοινὸν μέτρον ἐστίν. καὶ φανερόν, ὅτι καὶ μέγιστον· μεῖζον γὰρ τοῦ Δ μεγέθους τὰ A, B οὐ μετρεῖ.

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1. ἐστίν] om. F. μεῖζον] supra scr. m. 1 P. τι μεῖζον F, sed corr. 2. μεγέθη μετρήσει Theon (BFVb). τό] (alt.) m. 2 F. 3. ἔστί BVb, comp. F. 5. μέτρο P, sed corr. εὗρηται P. Deinde add. τὸ AZ V, sed punctis notat. 6. δεῖξαι] ποιῆσαι B et b (mg. γρ. δεῖξαι), δεῖ δεῖξαι F (mg. m. 2: γρ. ποιῆσαι). 9. μετρῇ] -η in ras. P. 15. Ante δεῖ ras. 1 litt. P. 16. δύο] om. V. 17. δή] m. rec. P. 18. μετρεῖ] om. P. 19. μετρεῖ δέ — 20. μετρεῖ] mg. m. 1 P. 20. Δ ἄρα] δὲ Δ P. τῶν] -ν postea add. F. ἐστί BV, comp. Fb. 21. καί] (alt.) om. BVb. 22. μέγεθος Fb. Post B ras. 1 litt. V. Post μετρεῖ add. εἰ γὰρ δυνατόν, μετρείτω τὰ A, B, Γ μεῖζον τοῦ Δ (μεγέθους add. V) τὸ E· καὶ ἐπεὶ τὰ A, B, Γ μετρεῖ,

tietur, maior minorem; quod fieri non potest. itaque magnitudo maior magnitudine AZ magnitudines AB, $\Gamma\Delta$ non metietur. ergo AZ magnitudinum AB, $\Gamma\Delta$ maxima mensura communis est.

Ergo datis duabus magnitudinibus commensurabilibus AB, $\Gamma\Delta$ maxima mensura communis inuenta est; quod erat demonstrandum.

Corollarium.

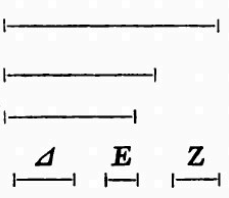
Hinc manifestum est, si magnitudo duas magnitudines metiatur, eandem maximam earum mensuram communem metiri.

IV.

Datis tribus magnitudinibus commensurabilibus maximam earum mensuram communem inuenire.

Sint datae tres magnitudines commensurabiles A, B, Γ. oportet igitur magnitudinum A, B, Γ maximam mensuram communem inuenire.

Sumatur enim duarum magnitudinum A, B maxima mensura communis [prop. III] et sit Δ. Δ igitur



magnitudinem Γ aut metitur aut non metitur. prius metiatur. iam quoniam Δ magnitudinem Γ metitur, et etiam A, B metitur, Δ magnitudines A, B, Γ metitur. Δ igitur magnitudinum A, B, Γ communis est mensura. et adparet, eandem maximam esse; nam magnitudo maior magnitudine Δ non metitur A, B.

καὶ τὰ A, B μετρήσει καὶ τὸ τῶν A, B μέγιστον κοινὸν (κοινὸν μέγιστον V) μέτρον τὸ Δ μετρήσει (μετρήσει τὸ Δ V) τὸ μεῖζον τὸ ἔλαττον (ἔλασσον V). ὅπερ ἄτοπόν ἐστιν (ἀδύνατον V) V et mg. m. 2 B.

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